Trading the Crude Curve

A primer on curve, relative-curve and lead-lag trading in WTI, Brent and the WTI-Brent arb, with notes on how prop shops run curve books

Internal brainstorming notes for the crude-quant-dashboard project · October 2026 · Educational material, not investment advice

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1. The building blocks

A futures curve has three things you can trade: its **level**, its **slope** and its **curvature**. Each standard structure is a fixed set of weights on consecutive contracts that cancels some of these and keeps the rest.

Structure	Weights (consecutive months)	What it isolates
Outright	`(1)`	Level: the price itself
Spread	`(1, −1)`	Slope: cancels level
Butterfly (fly)	`(1, −2, 1)`	Curvature: cancels level and (to first order) slope
Double-fly (dfly)	`(1, −3, 3, −1)`	A kink: one fly out of line with the next. Dfly(n) = Fly(n) − Fly(n+1)
Intermarket fly	`fly on (WTI − Brent)`	Whether WTI curvature differs from Brent curvature at the same point. 6 legs; the intermarket dfly is 8

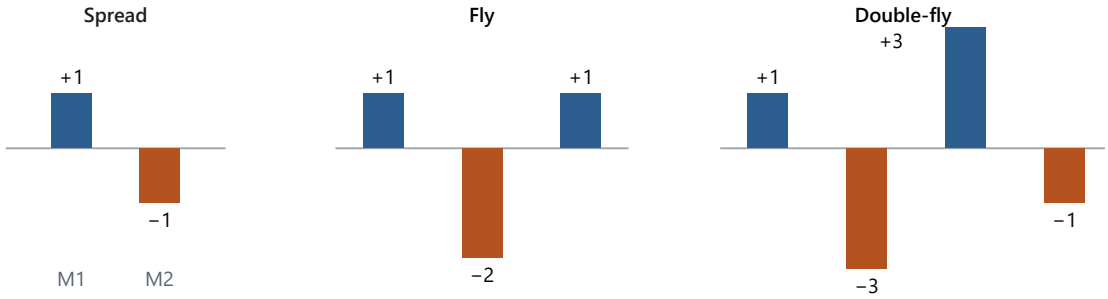


Figure 1. Leg weights. Blue legs are bought, orange legs sold. The weights in each structure sum to zero, so a parallel move in the curve cancels out.

> **SIGN CONVENTION**
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> Fly = A − 2B + C = (A + C) − 2B. A **negative** fly means the middle month B is **rich** (priced above the average of its wings). A positive fly means B is cheap. Buying a fly is buying the wings and selling the body.

Because every structure is a linear combination of outrights, you only need to store outright prices per contract. Flies, double-flies and intermarket structures can be recomputed on demand, which is also how the dashboard's structure engine prices them live.

2. Reading the fly curve

A fly-curve chart plots one fly per point along the curve. Each point is a fly of three consecutive months, labelled by its contract. It shows at a glance where the curve bends and where it is straight.

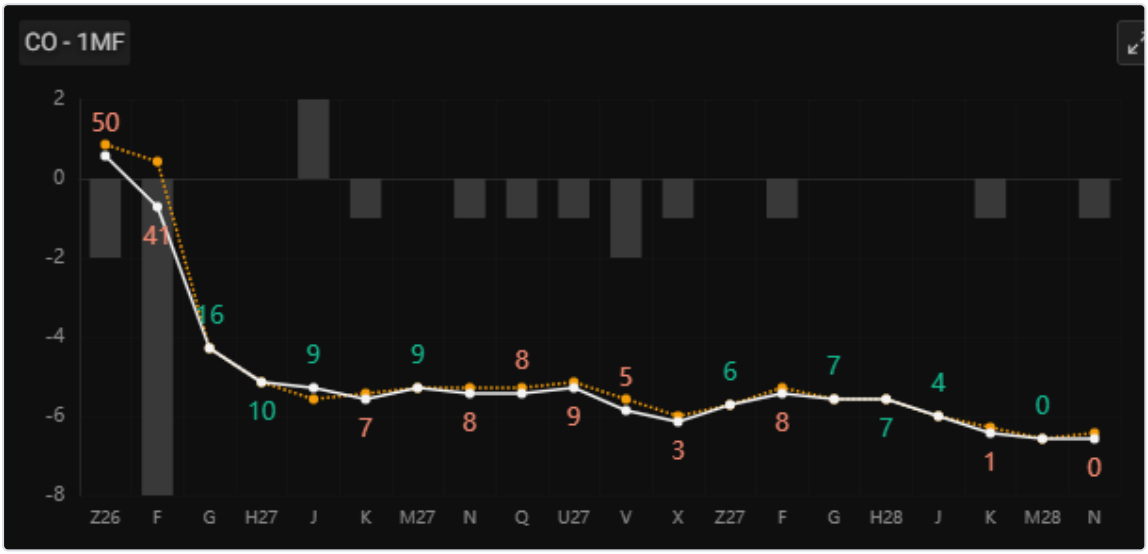


Figure 2. Brent (CO) 1-month fly curve from the dashboard. Read here as: white line = current fly values, orange dotted line = previous settle, grey bars = change. The red and green numbers are not interpreted in this document.

What this particular curve shows

- **Two regimes.** The front (Z26, F27, G27) sits well above the rest and drops steeply. From H27 onward the curve is a smooth, gentle decline from about −5 to −6.5. The front is driven by physical supply and demand; the back end is driven by hedging flows and risk premia.
- **Today's action is at the front.** The F27 fly has the largest change on the chart (the long grey bar). Front curvature is moving while the back end is barely changing.
- **Small kinks in the back.** The back end has small wiggles around K27 and V/X27. Each is a candidate double-fly: a fly that is out of line with its neighbours.

Smooth versus kinked

In a well-behaved market the fly curve is smooth, because nothing in fundamentals makes one deferred month special relative to the next. A kink is either a mispricing, or there is a reason for it (see Section 3). The double-fly measures exactly that kink.

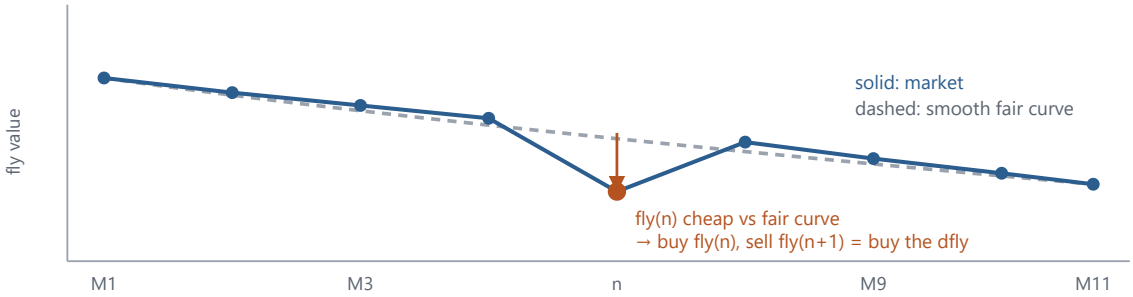


Figure 3. A kink on a fly curve. The double-fly around point n is the trade that profits if the kink closes back toward the smooth curve.

The same chart built on the per-month WTI − Brent spread is the **intermarket fly curve**. Everything above applies to it, with the extra expiry caveats covered in Section 4.

3. Curve trading: trade the shape, not the price

Curve trading means taking positions in slope and curvature while staying close to flat on outright price. The core idea is **smoothness**: the fly curve should be smooth, and a kink without a reason is something to fade. A double-fly is that trade in one instrument: buy the cheap fly, sell its rich neighbour.

There are four sources of edge. They are listed roughly from most to least reliable.

1. Flow

Someone has to trade a structure regardless of price, and you take the other side. Most curve P&L comes from knowing these flows, not from forecasting.

Flow	Effect on the curve
Producer hedging	Producers sell deferred contracts (often 12–24 months out), pushing the back end down and creating persistent pressure in specific months.
Consumer hedging	Airlines, refiners and shippers buy strips, often whole calendar years.
Bank swap hedging	Banks that sell swaps to clients hedge with calendar strips. Month-end and quarter-end can be heavy.
Index rolls	Commodity index funds roll from the front contract to the next, typically on business days 5–9 of the month (the "Goldman roll" window). This weighs on the front spread in a predictable window.
Expiry and options expiry	Positions are closed or rolled before last trade. Option-related hedging concentrates around strikes before options expire.

2. Relative value

Measure kinks against a fitted fair curve. This is the job of the PCA residuals and Nelson-Siegel residuals in the model plan: the residual says how far each fly or dfly sits from where a smooth curve says it should be.

3. Carry and roll-down

As a structure ages, it slides along the curve toward the front. If the shape of the curve stays the same, the structure's value changes simply by moving to a new point. Holding structures where that slide is in your favour earns carry.

4. Fundamentals, mostly at the front

- **The front is physical.** WTI front spreads respond to Cushing, Oklahoma stock levels. Brent front spreads respond to North Sea loading programmes and the Dated Brent physical market.
- **The back end is mostly financial.** It moves with hedging flows and risk premia.
- On the Brent chart in Section 2, the sharp drop between Z26/F27 and G27 is this physical-to-financial boundary showing up.

> **WHEN A KINK IS NOT A TRADE**
> Before fading a kink, check for a reason it should exist: an expiring contract, an index-roll window, the extra liquidity and open interest in December contracts, seasonal demand (gasoline season, winter heating), or a known large hedge. A kink with a reason can persist or get bigger.

4. Relative curve trading: one curve against another

Relative curve trading compares the *shape* of two related curves rather than one curve against itself. For crude, the main pair is WTI against Brent.

Why the WTI-Brent arb is anchored

Brent minus WTI is bounded by the cost of moving US crude to Europe: pipeline tariffs to the Gulf Coast, export-terminal costs and transatlantic freight. Since 2023, WTI Midland has also been one of the grades in the Dated Brent basket, which ties the two benchmarks together more directly. Because the level of the arb is anchored, the shape of the arb curve should be fairly stable.

What the intermarket fly tells you

An intermarket fly asks whether Brent's curvature differs from WTI's at the same point on the curve. Because the arb level is anchored, it is usually reacting to a local event:

- a North Sea outage or loading-programme change (Brent-specific)
- a Cushing build or draw (WTI-specific)
- one market's contract approaching expiry

> **EXAMPLE**
> The Brent F27 fly is down sharply on the day. If the matching WTI fly has not moved, the intermarket fly has moved. The question is then whether this is a Brent-specific story (which can persist) or a lag (which should converge as WTI catches up, or Brent reverts).

The expiry mismatch at the front

Contract	Last trading day
ICE Brent	Last business day of the second month before the delivery month
CME WTI	Three business days before the 25th calendar day of the month before the delivery month

So Brent month X expires roughly a month before WTI month X. The convention pairs the same named delivery month, which means front intermarket structures compare contracts with different times to expiry. They carry predictable distortions around Brent expiry; flag those dates rather than trading them as signals.

Settlement timing

Both contracts settle around 14:28–14:30 New York time, but ICE sets Brent's window in London time. In the weeks when US and UK daylight saving start or end on different dates, the windows slip apart. For intraday intermarket work, use synchronised price snapshots rather than relying on "the close".

Other relative-curve pairs

Pair	What drives it
Brent vs Dubai (EFS spread)	Light sweet vs medium sour crude; OPEC+ supply and Asian demand
Crude vs products (cracks)	Refining margins; RBOB, heating oil and gasoil curve shapes against crude
WTI Cushing vs Houston / Midland	US pipeline capacity and export demand

5. Lead-lag trading: who moves first

Lead-lag trading uses the fact that related instruments do not reprice at the same instant. It works at two very different speeds, and only one is realistic for a dashboard-driven trader.

Microstructure: milliseconds to seconds

- **Along the curve:** front outrights move first. Spreads, flies and back months reprice afterwards, often through *implied pricing*, where the exchange derives back-month prices from front outrights plus spread markets.
- **Across exchanges:** ICE Brent is matched in the UK and CME WTI in Illinois, so there is real transatlantic latency between them.
- **Who wins:** firms with co-location and microwave links. This cannot be captured from a dashboard. It does explain why back-month quotes sometimes look stale.

Statistical: minutes to days

- **Session leadership:** Brent tends to lead in London hours. WTI tends to lead in US hours, especially around the EIA weekly inventory report (Wednesdays, 10:30 New York time).
- **Flat-price beta:** when flat price rallies, front spreads usually strengthen and front flies move with them. A fly that lags its usual beta is a signal.
- **Error correction:** if WTI and Brent structures are cointegrated (tied together over time), the size of the gap predicts which one moves back.

How to test for lead-lag

Method	What it answers
Lagged cross-correlation	Do changes in A today correlate with changes in B over the next 1, 5 or 15 minutes (or days)?
Granger causality	Do A's past changes improve a forecast of B beyond B's own history?
Vector error-correction model (VECM)	When A and B drift apart, which one does the adjusting, and how fast?
Rolling beta to front spread / flat price	How much should a fly have moved given what the liquid part of the curve did?

> **HOW TO USE IT**
>
> At non-HFT speeds, treat lead-lag as a **timing tool** for entering trades the relative-value view already likes, not as a standalone strategy. Leadership also shifts by session and regime, so re-estimate it regularly instead of assuming one market always leads.

6. How prop shops trade the curve

This section describes general, publicly discussed industry practice, not the methods of any specific firm.

Market-make around a fair-value curve

Desks quote spreads and flies on both sides around a model fair-value curve. The curve is anchored on the most liquid points (front outrights and listed spreads), and illiquid months are priced from the model. They skew their prices to manage inventory and earn the bid-ask spread plus flow. Prediction is secondary.

Manage risk in factor space

Positions across hundreds of legs are netted into level, slope and curvature exposures, using PCA-style factors, then hedged with the cheapest liquid instruments. A trader can be long 40 different back-month flies and still be close to flat in factor terms. What matters is the net factor exposure and the residual, not the gross position count.

Assign traders to curve sections

Section	Character
Front spreads	Physical and fundamentals-driven: inventories, loadings, refinery runs, expiry dynamics
Deferred curve	Flow and strip-driven: producer and consumer hedges, bank swap hedging
Intermarket arb	Its own book: freight, export capacity, Midland in Dated Brent, expiry mismatch

Section owners know who is hedging what and when, which is often worth more than any model signal.

Leg in passively

Double-flies are rarely listed, so they are built from legs. The usual approach is to rest an order on the illiquid leg and hedge the liquid legs aggressively when it fills. Execution quality is a large part of the edge in dflies, because a few ticks of slippage on four legs can wipe out the expected reversion.

Keep an event playbook

Known patterns around expiry, the index-roll window, EIA reports, OPEC+ meetings and month-end strip hedging are written down and traded repeatedly.

Run high-frequency lead-lag separately

The latency-sensitive lead-lag business sits in its own engine with its own infrastructure, usually apart from the discretionary curve desks.

7. What this means for the dashboard

The planned PCA and Nelson-Siegel work fills the **relative-value fair-curve** role from Section 3. To make it tradable the way a desk would use it, three additions matter most.

Addition	Why
Flow and event calendar	Expiries for both markets, the index-roll window, EIA, OPEC+ meetings and month-end strip-hedging windows. This tells you when a kink has a reason and should not be faded.
Intermarket overlay	The same fly curve for WTI, Brent and the arb on one chart, plus each fly's beta to the front spread. This makes relative-curve and lead-lag moves visible at a glance.
Factor-exposure risk	Net level, slope and curvature exposure of the positions, so a book of flies is checked for accidental slope or level risk.

Mapping ideas to models

Trading idea	Model or data that supports it
Fade kinks (dfly)	PCA residual z-score; Nelson-Siegel residual per contract
Carry / roll-down	Fitted curve; expected fly value after one roll period
Flow timing	Event calendar; contract calendar with expiries and roll windows
Intermarket divergence	Separate WTI and Brent fits; cointegration / error-correction on the arb
Entry timing	Statistical lead-lag; z-score, RSI, OU half-life on synchronised structure series
Avoid trends	ADX or Hurst exponent as a "don't fade this" filter

Glossary

Arb	The WTI − Brent spread for the same delivery month
Backwardation / contango	Front months above deferred months / below deferred months
Cointegration	Two series that can drift apart in the short run but are tied together over time
Implied pricing	Exchange-derived prices for one market from the prices of related outright and spread markets
Index roll	Commodity index funds moving positions from the expiring contract to the next one, typically business days 5–9
Kink	A fly out of line with its neighbours on the fly curve; measured by the double-fly
Roll-down	The change in a structure's value as it ages along an unchanged curve
Strip	A set of consecutive months traded together, e.g. a full calendar year

Contract rules (last trading days, settlement windows, basket composition) should be checked against current ICE and CME rulebooks before they are coded into the dashboard.